Effects of Frontal tDCS Applications on Cognitive Functions in Healthy Individuals

İbrahim Sarı, MD, MSc[1], Aslı Demirtaş-Tatlıdede, MD, PhD[1,2]

[1]BAU Health Sciences Institute, Neuroscience PhD Program, Istanbul
[2]BAU Faculty of Medicine, Department of Neurology, Istanbul

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Introduction

Transcranial direct current stimulation (tDCS) is a novel neuromodulation technique that is currently under research for modulation and enhancement of cognitive functions. Anodal tDCS is considered to induce plasticity and enhance the underlying cortical activity and the cathodal tDCS reduces it. While several studies suggest that one session of tDCS can modulate cognition in healthy individuals, available data is quite variable and further research is needed.

In this study, we aimed to explore the cognitive enhancing effects of tDCS applied over the dorsolateral prefrontal cortex (DLPFC) and medial prefrontal regions (PFC) in a group of healthy subjects.

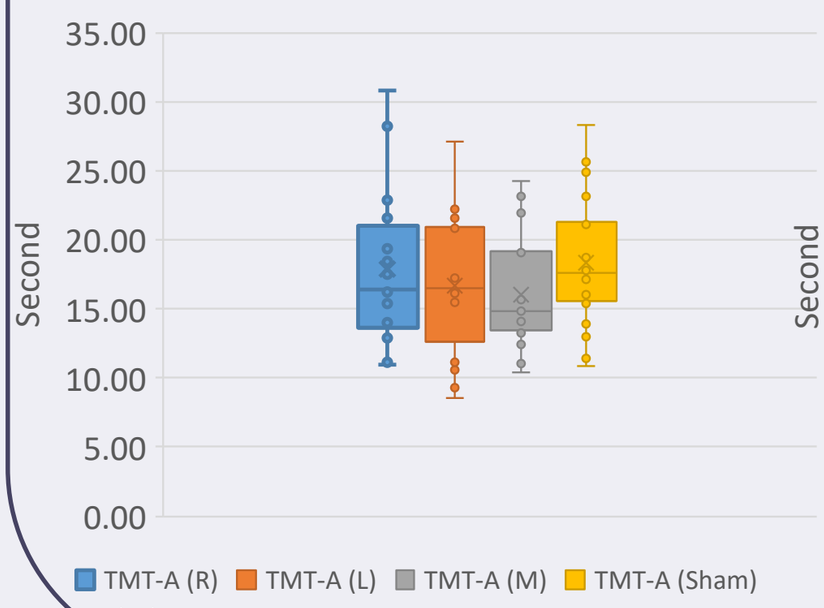
Materials and Method

In this randomized, within-subject, sham-controlled study, 20 healthy individuals received four courses of anodal tDCS stimulation (2mA) over the right DLPFC, left DLPFC, medial PFC, and sham, in four separate sessions. Participants needed to be aged between 18-45 years, capable of attending the clinic for tDCS sessions, no history of neurological and psychiatric disease and have received at least five years of formal education. The study protocol was approved by local ethics committee (Istanbul Medipol University, Ethics Committee for Non-interventional Clinical Studies), and the study was conducted in accordance with the declaration of Helsinki. All subjects provided informed consent prior to study entry and before the study procedures.

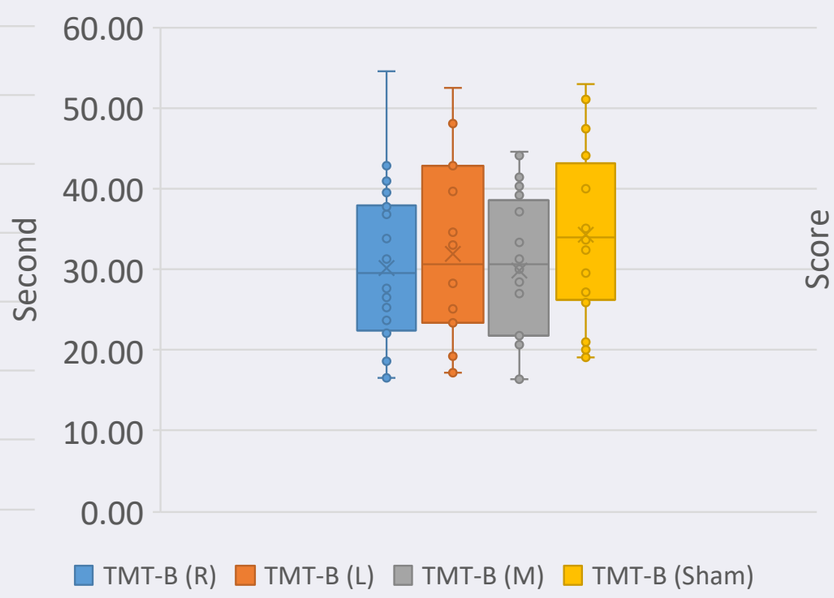
During the tDCS application, the following battery of tests was administered online to assess various dimensions of cognitive function: Trail Making Test (TMT-A and TMT-B), Letter-Number Sequencing (LNS), Verbal Fluency Test (VF), Symbol Digit Modalities Test (SD), Stroop Test, and Free and Cued Selective Reminding Test (FCSRT). The statistical analysis was carried out using SPSS software. Non-parametric analysis was performed due to the low sampling size. Friedman test with Wilcoxon signed-rank tests and Bonferroni correction for post-hoc analysis was used for within-group analysis.

Results

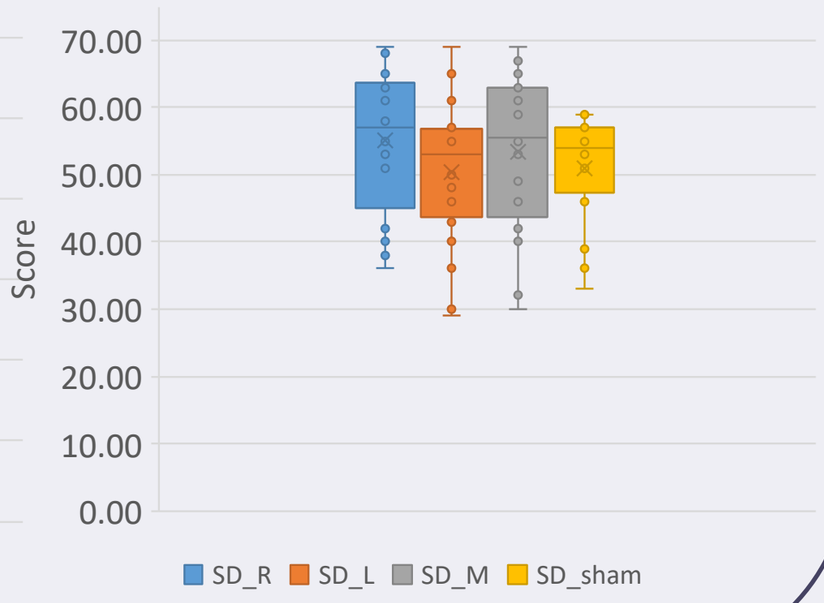
Active tDCS applications pointed to significant changes in several tests compared to sham tDCS. For the TMT-A test, significant improvements were observed in the medial PFC tDCS condition, with participants completing the task significantly faster (16.03±3.88 sec) compared to the sham condition (18.34±4.64 sec) ($p<0.001$). Similarly, the left DLPFC tDCS condition also showed **significantly** faster completion times (16.64±4.79 sec) than the sham condition ($p<0.001$). In the same direction, In the TMT-B test, the medial PFC tDCS condition demonstrated significantly faster performance (29.83±9.23 sec) compared to the sham condition (34.29±10.44 sec) ($p<0.001$). Participants under the right DLPFC tDCS condition (30.18±10.29 sec) also performed better than those under the sham condition, this difference was also statistically significant ($p<0.001$). Further, symbol digit test modalities test reveal a significant improvement compared to the sham condition when applied to right DLPFC (55.15±10.35 vs. 50.90±8.27) ($p<0.01$).



(A) Trail Making Test-A



(B) Trail Making Test-B



(C) Symbol Digit Modalities Test

Conclusions

Findings of this study highlight the clinical potential of tDCS in improving cognitive functions and processing speed in healthy objects via targeting the right, left DLPFC and medial prefrontal cortex. Medial prefrontal cortex may also be an important player for the attention and processing speed domains. Future studies will provide more support on the cognitive enhancing properties of tDCS in healthy individuals.

Contact: ibrahim.sari1@bahcesehir.edu.tr

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